

Documentation: As a technical report using demonstration of prototype (Sept 2026); external work additional to completion of study (August 2026), thesis submission (July 2026), Viva Voce exam passed (April 2026) and Defence Research Proposal (June 2024). Study commenced in December 2023. End of contract as Graduate Research Assistant on 30th March 2026.

Acknowledgement: The study uses Malaysian Agricultural Research and Development Institute (MARDI) data on the abundance and diversity of stingless bees in Malaysia (Jaapar et al. 2016) and the hourly satellite data from the POWER Project's Hourly v2.2.16 version accessed on 2024/02/10. Prediction of Worldwide Energy Resource (POWER) Project funded through the NASA Earth Science Division. Follows Bipartite Network Modelling Framework by Liew et al. (2023). Tools used include but not limited to: Jupyter Notebook, UCINET, VSCode, QGIS and Google Earth. Research conducted in enrolment to Faculty of Computer and Mathematical Sciences with Universiti Teknologi MARA, in fulfilment of the requirements for the degree of Master of Science (Information Technology), full-research mode; through the work as Graduate Research Assistant under the FRGS grant [FRGS/1/2023/ICT04/UITM/02/1] from Government of Malaysia. A patent registration has been approved through the University with the author accounting for 10% of the bipartite network model titled Diversity Abundance Bipartite Network (DivABNet) ownership. Bipartite Abundance Diversity (BAD) matrix is part of the network model; exclude the prototype version. Full text in thesis available from the University.

Title: Bipartite Network With Heterogeneous Features For Modelling Stingless Bee Preferred Habitat

Introduction:

The bipartite network approach is built on graph theory. The approach to pinpoint cause-and-effect was commonly simulated by trial-and-error in the past. Nowadays, machine automated processes and network modelling frameworks are able to minimise the trial-and-error. Thus, it is recognised that graph theory is widely used in explaining real-life phenomena network problem.

Objective:

1. To characterise graph structures underlying the habitat preference of stingless bees
2. To model the preferred habitat of stingless bees through bipartite network approach by incorporating the physical characteristics and environmental properties of every node
3. To evaluate the formulated bipartite preferred habitat network model of stingless bees by verification and validation

Scope:

The network model of habitat suitability for thirty-five species of stingless bees across 12 genera at 12 locations in Malaysia, where maximum body lengths range between 3 mm and 9 mm. In this study, habitat is defined as the location that can provide resources for optimum survival and reproduction of the stingless bees or synonymously, meliponine. The location nodes preferred by meliponine refer to an area of radius 500 m for every sampling site. The habitat suitability refers to the preferred habitat, which is presented in terms of the ranking of location nodes. The higher-ranked location nodes indicate relatively more preferred habitat by the meliponine and thus better suitability to function as the habitat of meliponine.

Significance of study

- Incorporating the heterogeneity nature of every bipartite node, and capturing the characteristics of the nodes at an individual level, for the species of individual stingless bees and the locations they are found.
- Policymakers in the agriculture ministry may discover the allocation of resources with the application of computer science through the demonstration of network studies.
- Potential to be helpful for guiding key decisions in urban forestry management, by identifying strategic locations based on species occurrence data and mapping of potential preferred habitat hotspots for stingless bees. This finding may be beneficial to the community of meliponine enthusiasts and bee apiary management offices.
- Potential to model effectiveness in researching ecological interactions through reproducibility and modifications within the bipartite network modelling framework, using expandable library of quantification techniques.
- This study uses random forest as the chosen machine learning technique, to quantify the weather conditions and physical characteristics of a location in relation to the abundance and diversity of stingless bees.
- The quantification of location characteristics and parameters related to bee species occurrence would contribute to the body of knowledge in apiology, that is the scientific study of honeybees; in category of pollinator insect species.
- Expressing attention for bees, to help creating bee-friendly agricultural landscapes, reducing reliance on chemicals in artificial pollination to foster more sustainable and organic farming practices. Future studies may advance to discover the visual acuity and heat-stress tolerance levels of bees, as these specific traits determine their capacity to forage and sustain healthy colonies.
- Potential to promote other benefits of a sustainable present-day ecosystem and natural habitats are preserved with bees in harmony with other species of life forms, when an approach of an enumerated inventory management technique is introduced; for planning recovery from disaster, food web management, prey-predatory management and rehabilitation of modern-day landscapes to rehome bees.

Findings:

- The centrality metrics calculated are degree centrality (DC), betweenness centrality (BC), eigenvector centrality (EC), and closeness centrality (CC): The findings in all four network metrics consistently reveal the same locations (L2, L3, L7, L8) and genera (M1 Heterotrigona, M2 Geniotrigona, M3 Lepidotrigona, M8 Tetragonilla, M9 Tetragonula) ranked highest.
- In the prototype, the ranked locations are revealed in the same order through power iteration method (Hyperlink-Induced Topic Search, HITS algorithm). However, the top ranked bee genus from prototype run is M9 Tetragonula, which has the highest number of species while the reported top ranked bee genus was M2 Geniotrigona which has the largest body measured in maximum body length. (Differences attributed to quantification technique deployed in September 2026 for the prototype, through development of the Jupyter Notebook first producing the *Wa* and *Wd* variables in January 2026).
- The locations preferred by the meliponines appear to be in the areas known to be preserved forests with water bodies and maintained recreation areas. L2 & L3 are located in States Perlis and Perak respectively. While L7 & L8 are located in States Terengganu and Kelantan. L7 is a man-made botanical garden for stingless bees, near to the largest man-made lake in Malaysia.

Link Matrix Using Reported Wa Wd results

		M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
1	L1	0.5	0.5	0.5	0.0	0.0	0.5	0.5	0.5	0.5	0.0	0.0	0.5
2	L2	0.7	0.7	0.7	0.7	0.7	0.8	0.7	0.7	0.7	0.7	0.7	0.7
3	L3	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
4	L4	0.5	0.5	0.5	0.0	0.5	0.5	0.5	0.5	0.5	0.0	0.0	0.0
5	L5	0.6	0.6	0.6	0.0	0.6	0.6	0.6	0.6	0.6	0.6	0.0	0.0
6	L6	0.7	0.7	0.7	0.0	0.0	0.7	0.7	0.7	0.7	0.7	0.7	0.0
7	L7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.0	0.0
8	L8	0.6	0.6	0.6	0.0	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.0
9	L9	0.8	0.8	0.8	0.0	0.0	0.8	0.8	0.8	0.8	0.0	0.8	0.0
10	L10	0.5	0.5	0.5	0.0	0.0	0.5	0.5	0.5	0.5	0.0	0.5	0.0
11	L11	0.5	0.5	0.5	0.0	0.5	0.5	0.0	0.5	0.5	0.5	0.0	0.0
12	L12	0.7	0.7	0.7	0.0	0.0	0.0	0.0	0.7	0.7	0.0	0.0	0.0

Link Matrix Using Prototype Wa Wd results

	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
L1	0.5	0.5	0.5	0.0	0.0	0.5	0.5	0.5	0.5	0.0	0.0	0.5
L2	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
L3	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
L4	0.5	0.5	0.5	0.0	0.4	0.5	0.4	0.5	0.5	0.0	0.0	0.0
L5	0.6	0.6	0.6	0.0	0.6	0.6	0.6	0.6	0.6	0.6	0.0	0.0
L6	0.7	0.7	0.7	0.0	0.0	0.7	0.7	0.7	0.7	0.7	0.7	0.0
L7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.0	0.0
L8	0.6	0.6	0.6	0.0	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.0
L9	0.8	0.8	0.8	0.0	0.0	0.7	0.7	0.7	0.8	0.0	0.7	0.0
L10	0.5	0.5	0.5	0.0	0.0	0.5	0.5	0.5	0.5	0.0	0.5	0.0
L11	0.5	0.5	0.5	0.0	0.5	0.5	0.0	0.5	0.5	0.5	0.0	0.0
L12	0.7	0.7	0.7	0.0	0.0	0.0	0.0	0.7	0.7	0.0	0.0	0.0

Wa Wd Score comparisons

	Wa Score	Prototype Wa	Reported Wa
L1		0.384579	0.327782
L2		0.545910	0.573756
L3		0.523344	0.573756
L4		0.229409	0.272891
L5		0.253201	0.213087
L6		0.418031	0.525049
L7		0.331454	0.368552
L8		0.297436	0.319844
L9		0.384579	0.317545
L10		0.516365	0.362260
L11		0.177298	0.166134
L12		0.210750	0.180276
	Wd Score	Prototype Wd	Reported Wd
L1		0.482267	0.499039
L2		0.633363	0.631087
L3		0.654467	0.631087
L4		0.367291	0.408396
L5		0.278058	0.312913
L6		0.382091	0.588719
L7		0.375913	0.500050
L8		0.267115	0.499039
L9		0.493210	0.351898
L10		0.654467	0.394265
L11		0.497045	0.263627
L12		0.438237	0.312913

Result of Ranking Algorithm Locations

L3	0.461191			L3	0.4633		
L2	0.378374			L2	0.3801		
L7	0.343114			L7	0.3435		
L9	0.311166			L9	0.3098		
L6	0.301224			L6	0.3011		
L8	0.289105			L8	0.2899		
L5	0.25004			L5	0.2497		
L10	0.219289			L10	0.2182		
L1	0.198394			L1	0.1968		
L11	0.192243			L11	0.1913		
L4	0.188187			L4	0.1873		
L12	0.18146			L12	0.1783		
HUB - LOCATIONS				HUB-LOCATIONS		k=5'	
PROTOTYPE RESULTS				REPORTED RESULTS OF 6.D.P USAGE			
k=5'				THIS ONE IS SENT FOR COMPLETING THE STUDY			
	6 D.P.						
ERROR	7.83E-08			ERROR:	9.71E-08		

Result of Ranking Algorithm on Meliponines

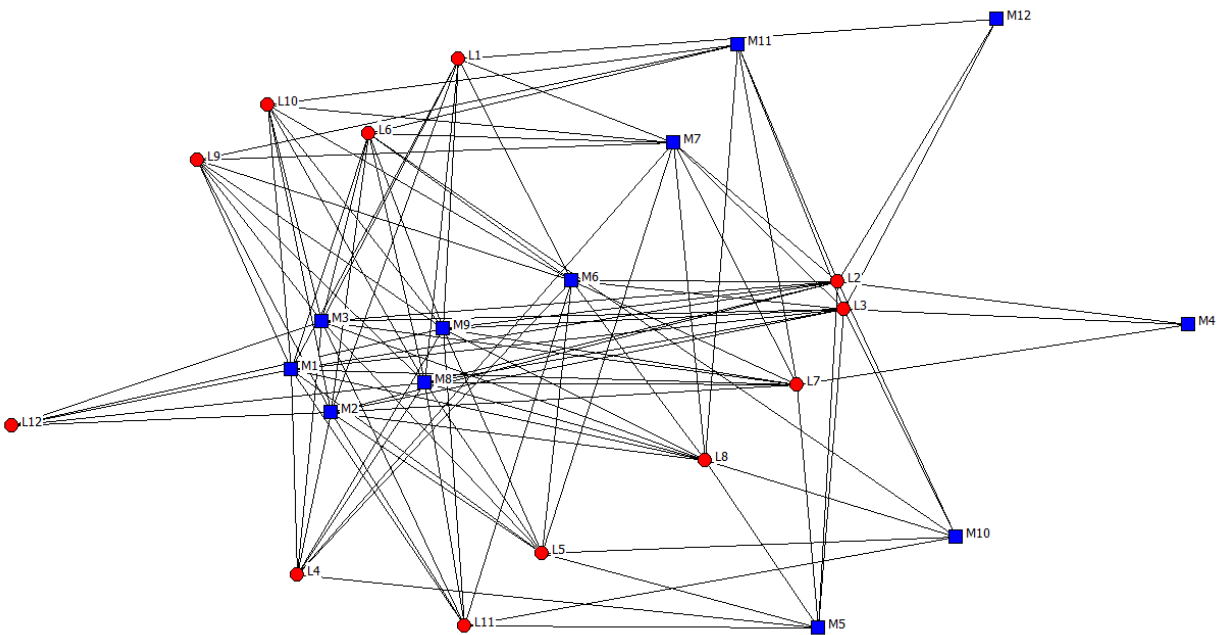
M9	0.352769			M2	0.3466		
M1	0.350276			M1	0.3461		
M2	0.348964			M8	0.3455		
M3	0.348373			M9	0.3451		
M8	0.343502			M3	0.3448		
M6	0.328463			M6	0.3282		
M7	0.308079			M7	0.312		
M11	0.254952			M11	0.2611		
M5	0.219618			M5	0.2251		
M10	0.200155			M10	0.2048		
M4	0.144783			M4	0.1472		
M12	0.12112			M12	0.1233		
AUTH - MELIPONINES (BEE)				AUTH - MELIPONINES (BEE)			
PROTOTYPE RESULTS				REPORTED RESULTS OF 6.D.P USAGE			
k=6'				k=6'			
	6 D.P.				6 D.P.		
ERROR	3.78E-10			ERROR:	4.42E-10		

References

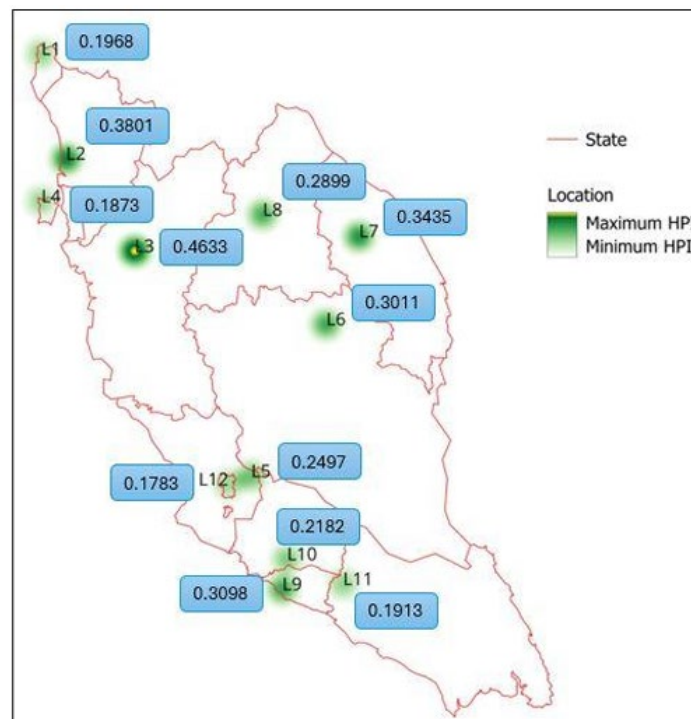
- Basari, N., Ramli, S., & Mohd Khairi, N. (2018). Food Reward and Distance Influence the Foraging Pattern of Stingless Bee, *Heterotrigona itama*. *Insects*, 9(4), 138. <https://doi.org/10.3390/insects9040138>
- Benedick, S., Gansau, J. A., & Ahmad, A. H. (2021). Foraging Behaviour of *Heterotrigona itama* (Apidae: Meliponini) in Residential Areas. *Pertanika Journal of Tropical* <https://doi.org/10.47836/pjtas.44.2.13>
- Borgatti, S.P., Everett, M.G. and Freeman, L.C. 2002. Ucinet 6 for Windows: Software for Social Network Analysis. Harvard, MA: Analytic Technologies.
- Harary, F. (1969). Graph Theory. Addison-Wesley Publishing Company.
- Jaapar, M. F., Halim, M., Mispan, M. R., Jajuli, R., Saranum, M. M., Zainuddin, M. Y., Ghazi, R., & Abd Ghani, I. (2016). The Diversity and Abundance of Stingless Bee (Hymenoptera: Meliponini) in Peninsular Malaysia. *American-Eurasian Network for Scientific Information*, 10(9), 1–7.
- Ng, W.J., Sit, N.W., Ooi, P. A.C., Ee, K.Y., & Lim, T.M. (2021). Botanical Origin Differentiation of Malaysian Stingless Bee Honey Produced by *Heterotrigona itama* and *Geniotrigona thoracica* Using Chemometrics. *Molecules*, 26(24), 7628. <https://doi.org/10.3390/molecules26247628>
- Liew, C. Y. (2016). Bipartite-Network-Based Modeling of Habitat Suitability (Doctoral dissertation, Universiti Malaysia Sarawak). UNIMAS Institutional Repository. <https://ir.unimas.my/id/eprint/24874>
- Liew, C. Y., Labadin, J., Kok, W. C., & Eze, M. O. (2023). A methodology framework for bipartite network modeling. *Applied Network Science*, 8(1), 6. <https://doi.org/10.1007/s41109-023-00533-y>
- Podani, J. The wonder of the Jaccard coefficient: from alpine floras to bipartite networks. (2021). *Flora Mediterranea*, <https://doi.org/10.7320/FIMedit31SI.105>
- Probst, P., Wright, M. N., & Boulesteix, A. (2019). Hyperparameters and tuning strategies for random forest. *WIREs Data Mining and Knowledge Discovery*, 9(3). <https://doi.org/10.1002/widm.1301>
- Roubik, D. W. (2006). Stingless bee nesting biology. *Apidologie*, 37(2), 124–143. <https://doi.org/10.1051/apido:2006026>
- Samsudin, S. F., Mamat, M. R., & Hazmi, I. R. (2018). Taxonomic study on selected species of stingless bee (Hymenoptera: Apidae: Meliponini) in Peninsular Malaysia. <https://api.semanticscholar.org/CorpusID:92795216>

NOTE: FULL LIST IN THESIS AVAILABLE FROM UNIVERSITY REPOSITORY.

APPENDIX



Network model visualised on NetDraw UCINET



Visualisation of locations using QGIS and their resulting Habitat Preference Indicator

Node Assignment	Location Name (Jaapar et al., 2016)	Coordinates	Nearest River
L1	Perlis: Perlis State Park	6.645111, 100.203389	Sungai Bawah Tanah Gua Kelam
L2	Kedah: Titi Hayun Recreational Forest	5.794389, 100.401722	Sungai Titi Hayun
L3	Perak: Lata Kekabu Recreational Forest	5.044417, 100.949417	Sungai Perak
L4	Penang: Kampung Tengah Teluk Bahang	5.448583, 100.203556	Sungai Teluk Bahang
L5	Selangor: Gunung Nuang Recreational Forest	3.217361, 101.882444	Sungai Pangsun
L6	Pahang: Kampung Pagi, National Park	4.450194, 102.482339	Sungai Pengau
L7	Terengganu: Pulau Tekak, Tasik Kenyir	5.164306, 102.744944	Sungai Kenyir
L8	Kelantan: Jelawang	5.339417, 101.974528	Sungai Kenerong
L9	Melaka: Sungai Udang Recreational Forest	2.300083, 102.131139	Sungai Udang
L10	Negeri Sembilan: Gunung Datuk Recreational Forest	2.543583, 102.169167	Sungai Johol
L11	Johor: Gunung Ledang Recreational Forest	2.341417, 102.617389	Sungai Ayer Panas
L12	WPKL: Bukit Nanas Recreational Forest	3.150389, 101.701583	Sungai Gombak

Location node assignment

Node Assignment	Genus	Species
M1	<i>Heterotrigona</i>	S1 (<i>Heterotrigona itama</i>), S11 (<i>Heterotrigona erythrogastra</i>)
M2	<i>Geniotrigona</i>	S2 (<i>Geniotrigona thoracica</i>)
M3	<i>Lepidotrigona</i>	S4 (<i>Lepidotrigona terminata</i>), S7 (<i>Lepidotrigona ventralis</i>), S8 (<i>Lepidotrigona nitidiventris</i>), S35 (<i>Lepidotrigona trochanterica</i>)
M4	<i>Lophotrigona</i>	S18 (<i>Lophotrigona canifrons</i>)
M5	<i>Platrigona</i>	S23 (<i>Platrigona pendleburyi</i>)
M6	<i>Tetrigona</i>	S5 (<i>Tetrigona apicalis</i>), S6 (<i>Tetrigona melina</i>), S25 (<i>Tetrigona binghami</i>), S27 (<i>Tetrigona zucchi</i>), S28 (<i>Tetrigona peninsularis</i>)
M7	<i>Homotrigona</i>	S14 (<i>Homotrigona aliciae</i>), S21 (<i>Homotrigona fimbriata</i>)
M8	<i>Tetragonilla</i>	S9 (<i>Tetragonilla atripes</i>), S10 (<i>Tetragonilla collina</i>), S17 (<i>Tetragonilla irridipennis</i>), S22 (<i>Tetragonilla fuscibasis</i>), S29 (<i>Tetragonilla rufibasalis</i>)
M9	<i>Tetragonula</i>	S3 (<i>Tetragonula laeviceps</i>), S12 (<i>Tetragonula fuscobalteata</i>), S15 (<i>Tetragonula gressitti</i>), S16 (<i>Tetragonula reepeni</i>), S19 (<i>Tetragonula geissleri</i>), S26 (<i>Tetragonula pagden</i>), S30 (<i>Tetragonula minangkabau</i>), S31 (<i>Tetragonula melina</i>), S32 (<i>Tetragonula pagdeni</i>), S33 (<i>Tetragonula minor</i>), S34 (<i>Tetragonula pallidistigma</i>)
M10	<i>Sundatrigona</i>	S13 (<i>Sundatrigona moorei</i>)
M11	<i>Hypotrigona</i>	S24 (<i>Hypotrigona scintillans</i>)
M12	Other	S20 (<i>Platytrigona hobbyi</i>)

Meliponine node assignment

Ranking of Location Nodes

Ranking	Node	Type of Location (Jaapar et al. 2016)	Name of Park or Recreational Forest (ReF)	Habitat Preference Indicator
1	L3	Forested river	Lata Kekabu ReF, Perak	0.4633
2	L2	Major mountain range origin	Titi Hayun ReF, Kedah	0.3801
3	L7	Major mountain range origin	Pulau Tekak, Tasik Kenyir, Terengganu	0.3435
4	L9	Forested river	Sungai Udang ReF, Melaka	0.3098
5	L6	Major mountain range at centre	Kampung Pagi, National Park, Pahang	0.3011
6	L8	Major mountain range origin	Jelawang, Kelantan	0.2899
7	L5	Industrialised state	Gunung Nuang ReF, Selangor	0.2497
8	L10	Major mountain range at centre	Gunung Datuk ReF, Negeri Sembilan	0.2182
9	L1	Major mountain range origin	Perlis State Park	0.1968
10	L11	Industrialised state	Gunung Ledang ReF, Johor	0.1913
11	L4	Highly developed land	Kampung Tengah Teluk Bahang, Penang	0.1873
12	L12	Highly developed land	Bukit Nanas ReF, WPKL	0.1783

Ranking of location nodes

Ranking of Meliponine Nodes

Ranking	Meliponine	Genus	Meliponine Aggregate Indicator
1	M2	<i>Geniotrigona</i>	0.3466
2	M1	<i>Heterotrigona</i>	0.3461
3	M8	<i>Tetragonilla</i>	0.3455
4	M9	<i>Tetragonula</i>	0.3451
5	M3	<i>Lepidotrigona</i>	0.3448
6	M6	<i>Tetrigona</i>	0.3282
7	M7	<i>Homotrigona</i>	0.3120
8	M11	<i>Hypotrigona</i>	0.2611
9	M5	<i>Platrigona</i>	0.2251
10	M10	<i>Sundatrigona</i>	0.2048
11	M4	<i>Lophotrigona</i>	0.1472
12	M12	Genus - Other	0.1233

Ranking of meliponine nodes

SRCC on Shannon–Wiener (SW) with HPI

	SW	HPI	Rank SW	Rank HPI	DivABNet	b_{SW_HPI}	$[b_{SW_HPI}]^2$
L1	2.10	0.1968	8	9		-1	1
L2	2.56	0.3801	2	2		0	0
L3	2.64	0.4633	1	1		0	0
L4	1.74	0.1873	11	11		0	0
L5	2.37	0.2497	4	7		-3	9
L6	2.34	0.3011	5	5		0	0
L7	2.46	0.3435	3	3		0	0
L8	2.3	0.2899	6	6		0	0
L9	2.01	0.3098	9	4		5	25
L10	2.24	0.2182	7	8		-1	1
L11	1.75	0.1913	10	10		0	0
L12	1.24	0.1783	12	12		0	0

Sum of $[b_{SW_HPI}]^2 = 36$

$$\rho_{Loc} = 1 - \frac{6 \times 36}{12(12^2 - 1)} = 0.87412587$$

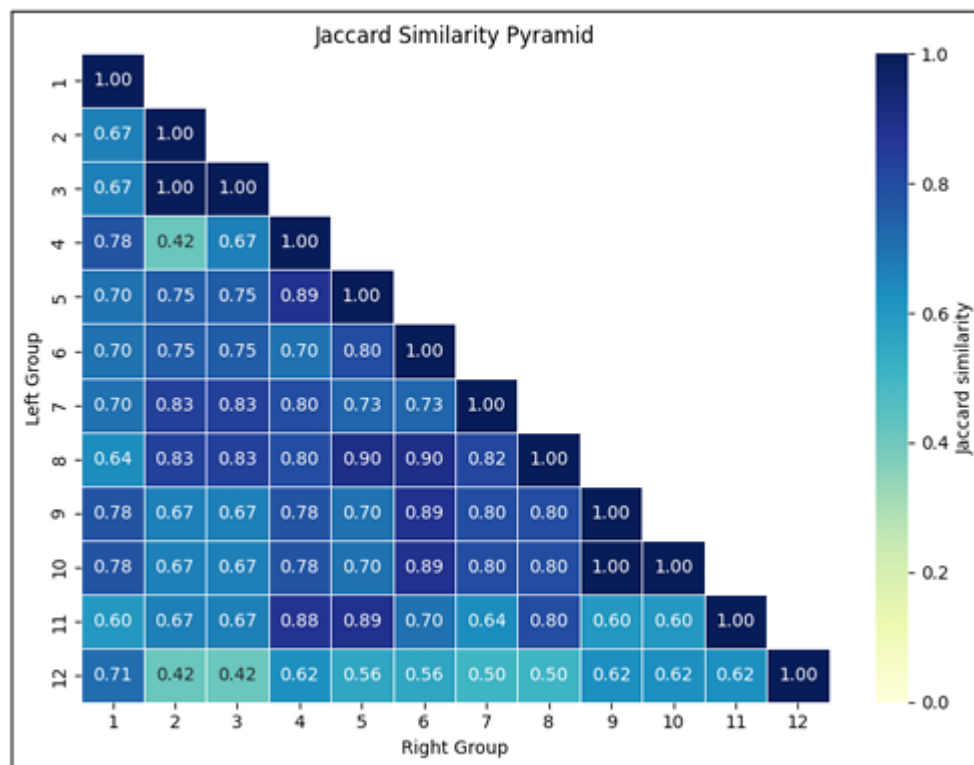
Validation of ranking of locations using Shannon-Wiener through SRCC computation

Location Node	Degree Centrality	Betweenness Centrality	Eigenvector Centrality	Closeness Centrality
L1	0.348	0.028041	0.185401	0.547619
L2	0.522	0.090227	0.239711	0.676471
L3	0.522	0.090227	0.239711	0.676471
L4	0.348	0.01532	0.194532	0.547619
L5	0.391	0.024806	0.208628	0.575000
L6	0.391	0.024944	0.208724	0.575000
L7	0.435	0.046487	0.218449	0.605263
L8	0.435	0.034169	0.225022	0.605263
L9	0.348	0.015092	0.194629	0.547619
L10	0.348	0.015092	0.194629	0.547619
L11	0.348	0.019415	0.185890	0.547619
L12	0.217	0.003294	0.130755	0.479167

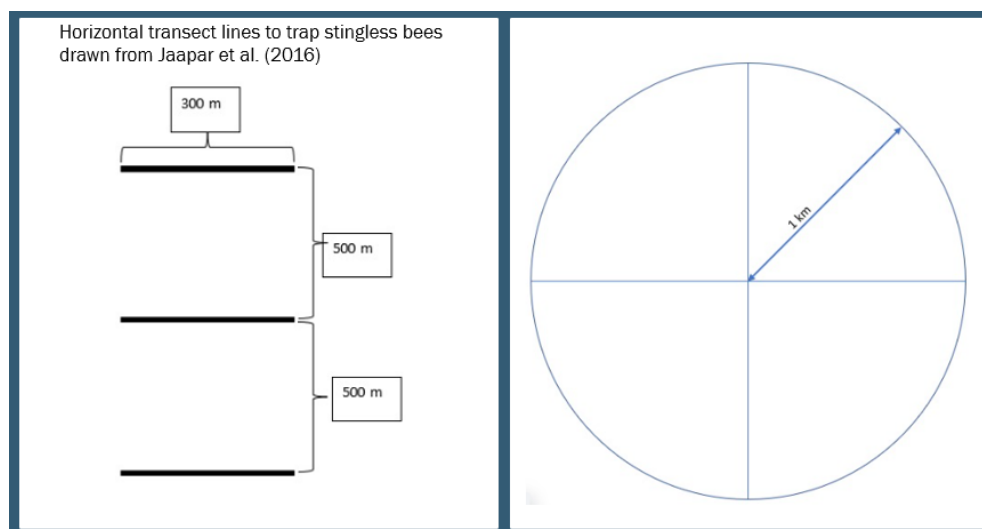
Network metrics calculated for location nodes

Meliponine node	Degree Centrality	Betweenness Centrality	Eigenvector Centrality	Closeness Centrality
M1	0.522	0.057099	0.251883	0.676471
M2	0.522	0.057099	0.251883	0.676471
M3	0.522	0.057099	0.251883	0.676471
M4	0.130	0.001565	0.072455	0.442308
M5	0.304	0.015252	0.156975	0.522727
M6	0.478	0.04287	0.238308	0.638889
M7	0.435	0.034328	0.219008	0.605263
M8	0.522	0.057099	0.251883	0.676471
M9	0.522	0.057099	0.251883	0.676471
M10	0.261	0.010517	0.135768	0.500000
M11	0.304	0.01493	0.157902	0.522727
M12	0.130	0.002158	0.069024	0.442308

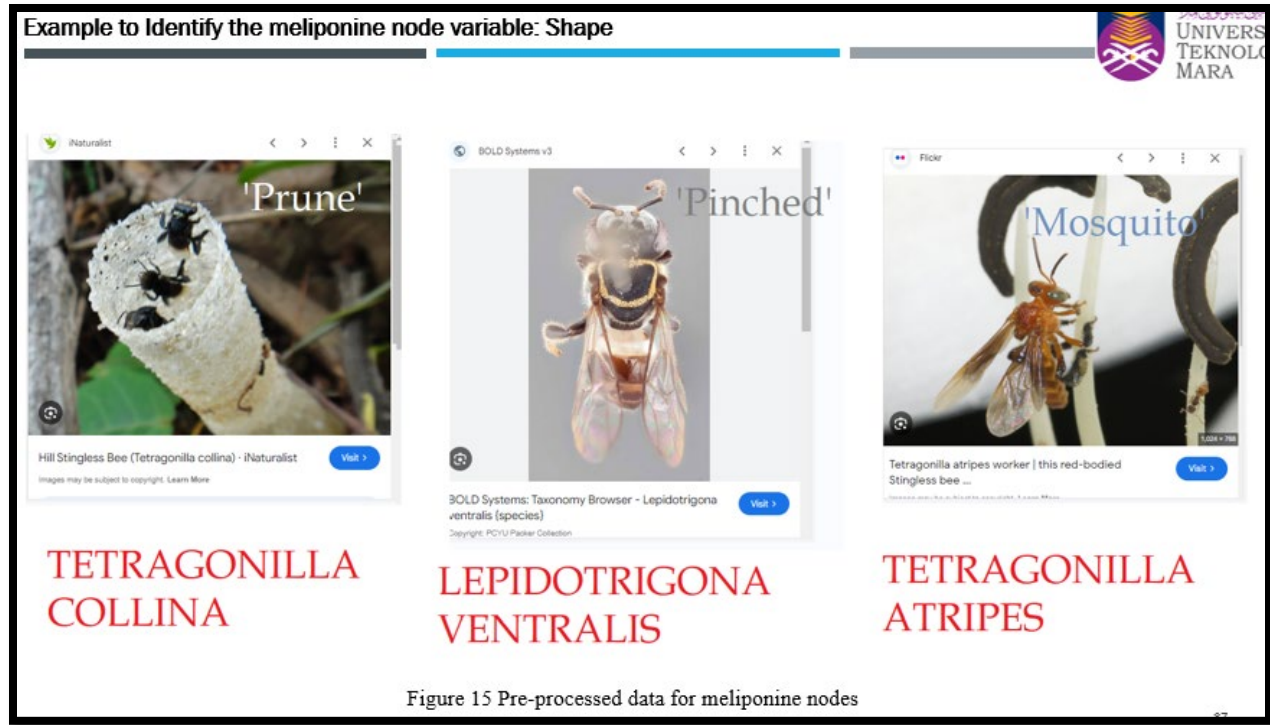
Network metrics calculated for meliponine nodes



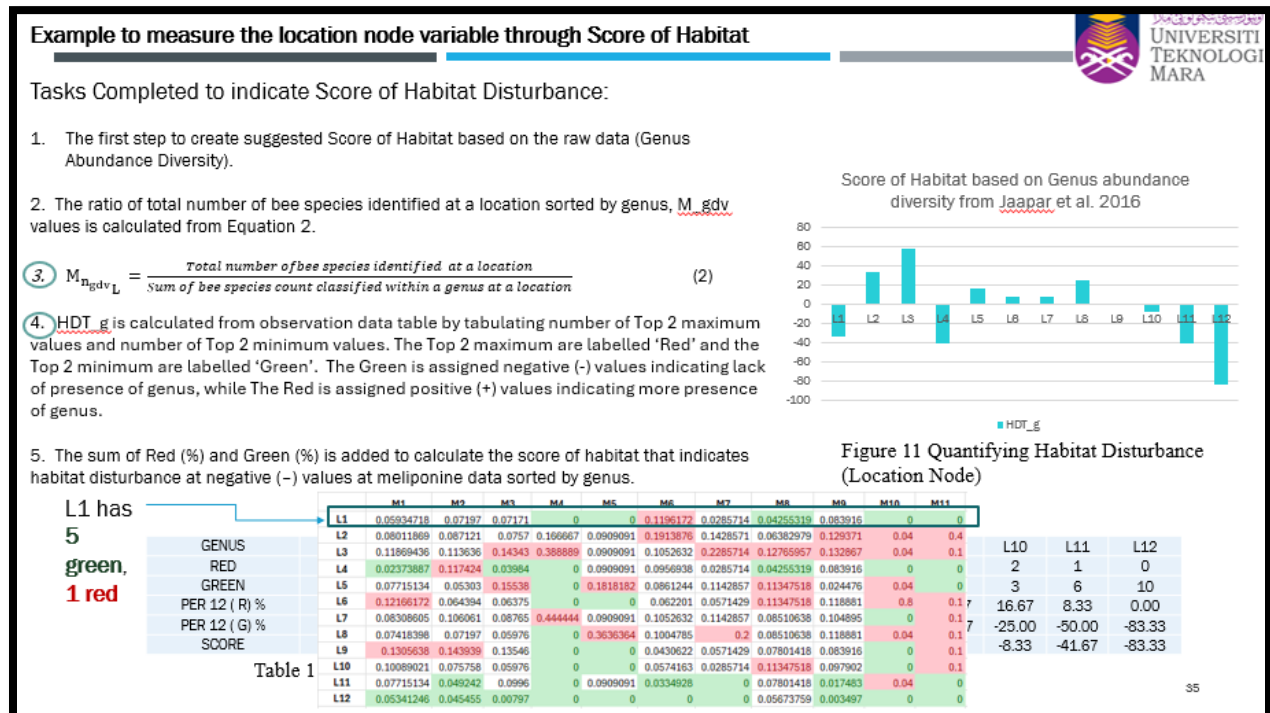
Jaccard Plot



Visualisation of trap



Previously archived identification techniques classify bees using object shapes that resemble the insects (Defence Research Proposal) excluded for starting with the base network model using limited morphometrics of the stingless bees through literature



Previously archived habitat disturbance score calculation method proposed (Defence Research Proposal) excluded for starting with the base network model implementation in the quantification of variables

Example to Identify the location node variable: Habitat Disturbance Type, HDT (Winfree et al. 2009)

Slide 50
Ctrl+Click to follow link

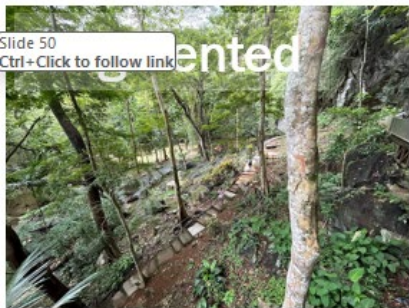


Figure 12 Sample HDT Fragmented



Figure 13 Sample HDT Logged



Figure 14 Sample HDT Agriculture

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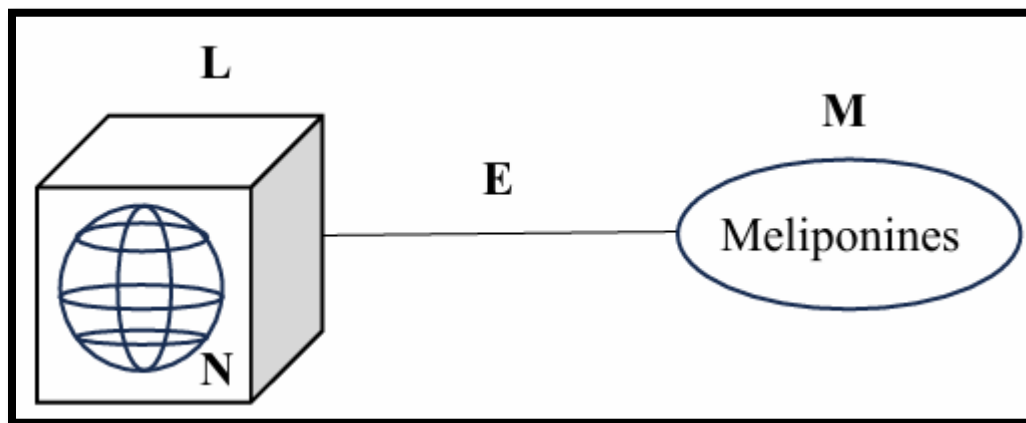
Previously archived location node classification method proposed (Defence Research Proposal) for others reference excluded for limited capability to acquire current state or state of habitat during sampling



Previously archived plot results using Orange showing location node effects grouped by types in weighted clusters of meliponine visualised based on relative humidity



Previously archived test run of HDT score vs. Diversity of Bees plot results using Orange (Defence Research Proposal) for reference



Basic building block of Diversity Abundance Bipartite Network (DivABNet)

$$\text{DivABG} = \text{DivABG}(L, M, E)$$

where $L = \{L1, \dots, L12\}$
 $M = \{M1, \dots, M12\}$
 $E = \{e1, \dots, e107\}$

Notation of Diversity Abundance Bipartite Graph (DivABG)